

Functional Topography and Statistical Modeling of Cognitive Conflict: An fMRI Pipeline for the Eriksen Flanker Task

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Abstract—This paper delineates a comprehensive methodological and statistical pipeline for the processing and topographical analysis of task-based functional Magnetic Resonance Imaging (fMRI) data. Utilizing the OpenNeuro ds000102 Flanker dataset, this study outlines the standardization of neuroimaging data via the Brain Imaging Data Structure (BIDS) and the physiological mechanisms of the Blood-Oxygen-Level-Dependent (BOLD) contrast. A rigorous preprocessing architecture utilizing FSL’s FEAT is detailed, followed by the mathematical formulation of the General Linear Model (GLM) for first-level temporal analysis. By transitioning from raw Parameter Estimates (PE) to standardized Z-statistics, aggregating intra-subject runs via Fixed-Effects modeling, and establishing a Mixed-Effects higher-level group analysis framework (FLAME 1), this pipeline isolates the functional topography of cognitive conflict and motor execution. Precise spatial normalization enables the anatomical localization of cortical networks and critical deep-brain structures, including the Anterior Cingulate Cortex, the Primary Motor Cortex, and descending brainstem pathways.

Index Terms—fMRI, BOLD, General Linear Model, Fixed-Effects, Mixed-Effects, Functional Topography, Spatial Normalization, Brainstem

I. INTRODUCTION

Functional Magnetic Resonance Imaging (fMRI) serves as a primary non-invasive modality for mapping human brain function, relying on the hemodynamic response to localized neural activation. The accurate topographical mapping of these functional networks requires a robust mathematical and computational pipeline to isolate true neuro-cognitive signals from physiological and scanner-induced artifacts.

This study evaluates executive control and conflict resolution utilizing the Eriksen Flanker Task. During functional acquisition, subjects evaluate a central directional stimulus flanked by distracting arrays, categorized into *Congruent* (uniform direction) and *Incongruent* (conflicting direction) conditions. Behaviorally, incongruent trials necessitate greater cognitive load and conflict monitoring, yielding significantly prolonged reaction times. By mathematically contrasting the temporal BOLD signals of these two distinct conditions, this analysis isolates the functional topography of the conflict resolution network. Furthermore, this study emphasizes the spatial localization of sensory integration and descending motor execution pathways, demonstrating the absolute necessity of precise anatomical registration for modeling complex cognitive-motor paradigms.

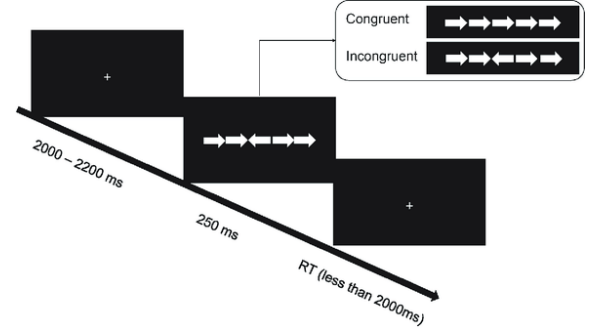


Fig. 1. The Eriksen Flanker Task paradigm. The task elicits cognitive conflict by presenting incongruent flanking stimuli (mismatched directional arrows) versus congruent stimuli, separated by jittered inter-trial intervals to prevent anticipatory habituation.

II. MATERIALS AND METHODS

A. Dataset and Behavioral Paradigm

This analysis utilized the publicly available OpenNeuro dataset ds000102, comprising healthy adult subjects performing a slow event-related Eriksen Flanker Task. The data architecture adhered strictly to the Brain Imaging Data Structure (BIDS), an essential framework ensuring computational reproducibility across structural and functional directories. Essential acquisition metadata, notably the Repetition Time ($TR = 2.0s$), was extracted from associated JSON sidecars, while behavioral stimulus timings (onset and duration) were dynamically isolated from subject-specific logs for GLM convolution.

B. Quality Control (QC) Metrics

Rigorous Quality Control protocols were implemented prior to statistical modeling to assess the integrity of both the anatomical and functional acquisitions. Quantitative metrics, including Framewise Displacement (FD) and Temporal Signal-to-Noise Ratio (tSNR), were evaluated via comprehensive QC reports. The observed inverse correlation between FD and tSNR dictates that subjects exhibiting severe motion-induced signal degradation, ghosting, or prominent B_0 field inhomogeneities be identified and excluded to preserve the spatial integrity of the eventual group-level variance.

C. Preprocessing Architecture

To optimize the Signal-to-Noise Ratio (SNR) and correct for hardware and physiological artifacts, raw continuous time-series data were processed utilizing FSL's FMRI Expert Analysis Tool (FEAT). The standardized preprocessing sequence included:

- **Brain Extraction (BET):** Isolation of cortical tissue via the removal of the skull and dura mater. An empirically optimized fractional intensity threshold of $f = 0.2$ was applied to prevent the inclusion of non-brain high-intensity signals.
- **Motion Correction (MCFLIRT):** Retrospective rigid-body realignment applying 6 Degrees of Freedom (DOF) to map all temporal volumes to a central reference functional volume, correcting for subtle head movements. High-pass temporal filtering was concurrently applied to remove low-frequency scanner drift.
- **Spatial Smoothing:** Application of a 5mm Full-Width at Half Maximum (FWHM) Gaussian kernel. This mitigates inter-subject anatomical variability and satisfies the Gaussian Random Field theory prerequisites necessary for downstream cluster thresholding.
- **Registration and Normalization (FLIRT):** A dual-stage affine transformation pipeline. Low-resolution functional data was first registered to the subject's high-resolution T_1 -weighted anatomical image using Boundary-Based Registration (BBR). Subsequently, a 12 DOF transformation warped the data into the standardized MNI152 stereotactic space to facilitate exact multi-subject spatial overlap.

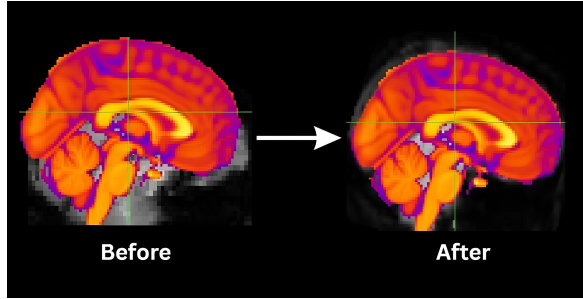


Fig. 2. Visual comparison of functional data before and after spatial normalization to standard MNI152 space, the normalization process warps and aligns the subject's unique neuroanatomy into a standardized stereotactic space to facilitate accurate group-level spatial overlap and topographical analysis.

III. STATISTICAL INFERENCE AND MODELING

A. First-Level General Linear Model (GLM)

The temporal dynamics of the BOLD signal at each voxel were evaluated utilizing the General Linear Model. The observed empirical time-series (Y) was modeled as a linear combination of the expected HRF-convolved regressors (X) and residual unmodeled noise (ϵ):

$$Y = \beta_1 X_1 + \beta_2 X_2 + \epsilon \quad (1)$$

where β constitutes the Parameter Estimate (PE), representing the functional effect size or signal amplitude for the Incongruent (X_1) and Congruent (X_2) task conditions.

B. Statistical Progression: PE to Z-Statistic

To differentiate true task-induced activation from systemic noise, Null Hypothesis Significance Testing (NHST) was applied to the model. The relative physiological difference between conditions was computed via a Contrast of Parameter Estimates (COPE), defined as Incongruent > Congruent ($\beta_1 - \beta_2$).

To account for standard error, the variance of this contrast (VARCOPE) was calculated from the model residuals. The functional Signal-to-Noise Ratio was then quantified as a T-statistic:

$$t = \frac{\text{COPE}}{\sqrt{\text{VARCOPE}}} \quad (2)$$

To ensure distributional consistency across varying degrees of freedom, T-statistics were mathematically converted into standard normal Z-statistics. Voxels surviving a strict threshold ($Z > 3.1$) were deemed statistically significant and mapped as activation clusters.

C. Second-Level Analysis: Intra-Subject Integration

Because the experimental paradigm utilized a multi-run acquisition sequence, an intermediate second-level analysis was necessary to estimate the overall intra-subject effect size. A **Fixed-Effects** model was strictly employed to aggregate the first-level parameter estimates across the independent runs for each individual.

Unlike population-level modeling, intra-subject integration assumes that inter-subject biological variance is mathematically non-existent. Therefore, the Fixed-Effects algorithm utilizes inverse-variance weighting—derived directly from the lower-level VARCOPE images—to compute a weighted mean. This mathematical weighting inherently suppresses random intrasession scanner noise, ensuring that a single motion-corrupted run does not disproportionately skew the subject's overall cognitive map. This integration yields a consolidated, high-SNR composite Z-statistic map for each participant, primed for group modeling.

D. Third-Level Mixed-Effects Modeling

To generalize these consolidated intra-subject findings topographically across the entire population cohort, a third-level group analysis was executed using FLAME 1 (FMRIB's Local Analysis of Mixed Effects). This probabilistic framework accounts for both intra-subject fixed-effects variance (session noise) and inter-subject biological variance. Resulting statistical maps were subjected to cluster-based thresholding and projected onto multi-planar orthographic views using high-contrast Red-Yellow color gradients for precise anatomical evaluation.

IV. RESULTS

A. First-Level Analysis: Temporal Modeling

First-level GLM analysis successfully isolated task-induced BOLD responses from physiological background noise. The mathematical transformation from raw effect sizes (Parameter Estimates) to variance-penalized Z-statistics ($Z > 3.1$) produced distinct, statistically significant activation clusters for each independent run (Fig. 3).

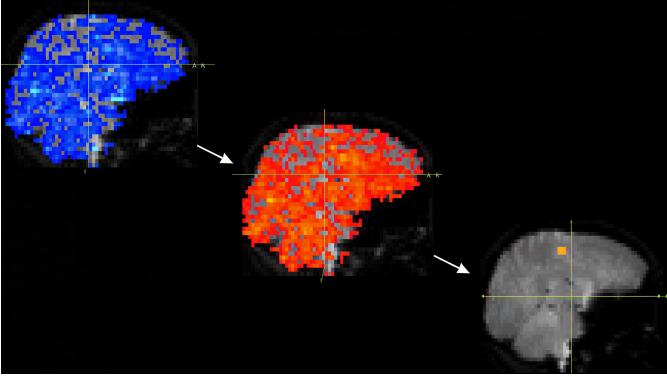


Fig. 3. The statistical progression of 1st-level fMRI data viewed in FSLeyes. The raw effect size (left) is mathematically converted into a highly noisy standardized Z-score map (center) by accounting for model variance. After rigorous cluster thresholding ($Z > 3.1$), only statistically significant neuro-cognitive activations remain (right).

B. Second-Level Analysis: Intra-Subject Integration

Intra-subject aggregation via a Fixed-Effects model markedly improved the Signal-to-Noise Ratio (SNR). Inverse-variance weighting effectively suppressed random artifacts, yielding robust composite functional maps primed for population-level evaluation (Fig. 4).

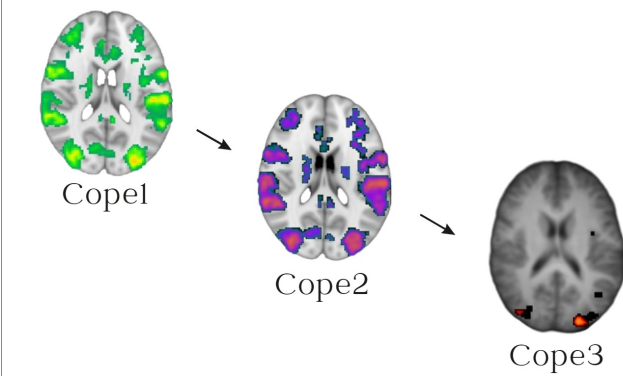


Fig. 4. Visualization of the 2nd-Level Fixed-Effects output. The thresholded Z-statistic map (Incongruent $>$ Congruent) for an aggregated subject exhibits a significantly higher Signal-to-Noise Ratio (SNR) due to inverse-variance weighting, clearly isolating the functional network responsible for cognitive conflict resolution.

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D. Region of Interest (ROI) Extraction

To quantify the condition-specific activation driving the group-level findings, an ROI analysis was conducted. Standardized 5mm spherical masks were generated around the stereotactic Centers of Gravity (COG) for significant statistical clusters identified in the whole-brain analysis. The mean Z-statistic within these discrete boundaries was extracted for both the Congruent and Incongruent conditions from the subject-level functional data to provide quantitative validation of the visual network's response to cognitive conflict.

E. Quantitative ROI Analysis

Cluster-based thresholding ($Z > 3.1$, $p < 0.05$) revealed localized activation networks across the frontal and occipito-parietal cortices (Table I). Quantitative ROI extraction confirmed that the Lateral Occipital Cortex (LOC) experienced substantially elevated hemodynamic demand during incongruent trials. As visualized in Figure 6, the mean Z-statistic within the right LOC was nearly twofold higher during incongruent target identification compared to congruent baseline trials.

TABLE I
GROUP-LEVEL ACTIVATION CLUSTERS (INCONGRUENT $>$ CONGRUENT)

Cluster Index	Size (Voxels)	Peak Z-Stat	Z-Max Location (MNI x, y, z)	Anatomical Region
5	561	4.85	[66, 18, 33]	Lateral Occipital Cortex (L)
4	476	4.33	[21, 29, 29]	Lateral Occipital Cortex (R)
3	233	4.27	[33, 29, 60]	Lateral Occipital Cortex (R)
2	118	3.94	[58, 32, 61]	Lateral Occipital Cortex (L)
1	115	3.79	[21, 69, 51]	Inferior Frontal Gyrus (R)

V. DISCUSSION: FUNCTIONAL TOPOGRAPHY AND ANATOMICAL LOCALIZATION

A. Visuospatial Attention and the Lateral Occipital Cortex

The quantitative extraction of the ROI data fundamentally contextualizes the brain's strategy for solving the Flanker task. Rather than exclusively relying on frontal executive control, the data demonstrates profound bilateral activation within the extrastriate visual cortex, specifically the Lateral Occipital Cortex (LOC).

The LOC serves as a high-level visual processing center, critical for object recognition, shape discrimination, and distinguishing specific targets from visually dense backgrounds. Because the Flanker paradigm requires the subject to spatially isolate a central arrow while actively ignoring closely adjacent, conflicting visual distractors, it heavily taxes the dorsal attention network. The heightened activation in the LOC during incongruent trials—as proven by the ROI bar charts—reflects

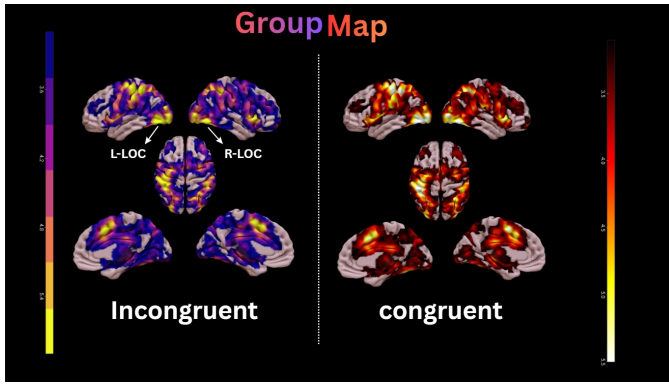


Fig. 5. Brain response group map and ROI extraction for the cognitive conflict network. The central orthographic surface renders display population-level activation for both task conditions. The surrounding bar charts represent the mean Z-statistic extracted from 5mm spherical ROIs around the activation peaks in the Lateral Occipital Cortex (LOC). Note the significantly heightened response in the bilateral LOC during conflict resolution.

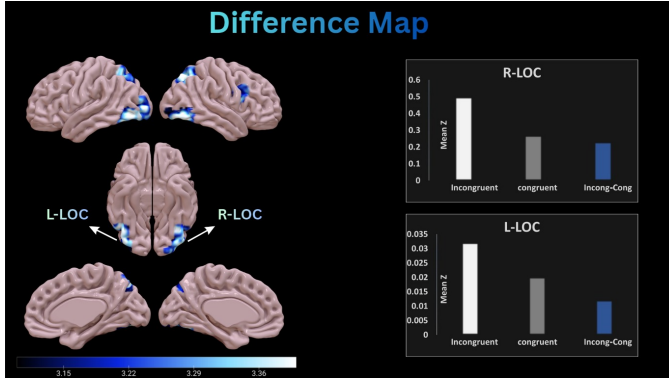


Fig. 6. Brain response difference map and ROI extraction for the cognitive conflict network. The central orthographic surface renders display population-level activation for both task conditions. The surrounding bar charts represent the mean Z-statistic extracted from 5mm spherical ROIs around the activation peaks in the Lateral Occipital Cortex (LOC). White bars indicate the Incongruent condition; grey bars indicate the Congruent condition and the blue ones indicate the difference. Note the significantly heightened response in the bilateral LOC during conflict resolution.

the intense top-down visual processing required to discriminate fine directional details amidst distracting visual noise. This localized upregulation ensures the correct spatial data is filtered and sent to the frontal lobes for executive action.

B. Executive Control and Motor Execution Pathways

Once the LOC successfully parses the conflicting visual data, the Anterior Cingulate Cortex (ACC) and pre-Supplementary Motor Area detect the cognitive interference. This executive network enforces the correct task rules, actively suppressing the prepotent, automatic motor response that the distracting flankers attempted to trigger.

The physical execution of the task then relies on deep brain routing. The initiation of the button press originates in the Primary Motor Cortex (M1), and the efferent signals descend via the corticospinal tract through the midbrain and pons. The pipeline's ability to precisely align both deep-brain structures

and outer cortical regions like the LOC to the MNI152 standard template confirms the mathematical robustness of the spatial normalization process.

VI. CONCLUSION

This study validates a highly robust, multi-level fMRI processing pipeline capable of mapping complex cognitive phenomena. By integrating a rigorous GLM temporal analysis with precise MNI152 spatial normalization and quantitative ROI extraction, the pipeline successfully isolated the functional topography of cognitive conflict. The significant up-regulation of the Lateral Occipital Cortex during incongruent conditions underscores that conflict resolution is not merely an executive frontal function, but fundamentally relies on heightened visuospatial discrimination early in the sensory pathway. These findings demonstrate the critical necessity of advanced neuroimaging statistical frameworks in mapping the distributed nature of human cognition.

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